

Research

Potential of quantum machine learning for solving the real-world problem of cancer classification

Mohadeseh Zarei Ghobadi¹ · Elaheh Afsaneh¹

Received: 21 June 2024 / Accepted: 17 September 2024

Published online: 27 September 2024

© The Author(s) 2024 **OPEN**

Abstract

Quantum machine learning (QML) algorithms have demonstrated the power of quantum computing for solving complex problems and big data in certain tasks. In this study, we explore the capabilities of QML for the classification of real-world biological large datasets including ten different cancer types based on gene expression values. By comparing the classification results obtained from the quantum algorithm with those from classical approaches, we disclose that the QML algorithm overall achieves comparable and reliable results. Moreover, we identify novel biomarkers that can contribute to the understanding of cancer biology. Some of these biomarkers are consistent with DNA promoter methylation. Our findings highlight the potential of QML in cancer classification and biomarker discovery, paving the way for future advancements in other disease research and clinical applications.

Article highlights

- QML could be implemented on real-world datasets to classify cancer types and identify biomarkers.
- QSVM outperformed some classical models in classification of ten cancer datasets.
- The novel biomarkers were found using quantum machine learning approach.
- Findings demonstrate the potential of QML in medical research and biomarker discovery.

Keywords Quantum machine learning · QML · Real-world problem · Classification · Cancer

1 Introduction

Quantum machine learning (QML) is an emerging and rapidly growing research field that lies at the intersection of quantum computing and machine learning (ML). In recent years, QML has attracted considerable attention due to its potential to enhance algorithms in certain tasks for faster problem-solving, higher accuracy achievement, and lower energy consumption. Quantum computing based on the principles of quantum mechanics such as superposition and entanglement, provides enhancements over traditional computing capabilities [1–3]. Due to the property of superposition, quantum bits (qubits) can exist in multiple states simultaneously, allowing quantum computers to process a vast

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s42452-024-06220-6>.

✉ Mohadeseh Zarei Ghobadi, mohadesehzaree@gmail.com; ✉ Elaheh Afsaneh, elahehafsaneh@gmail.com | ¹Independent Researcher, Tehran, Iran.



number of possibilities at once. This capability enables them to explore numerous combinations and find solutions much faster for specific problems. In addition, quantum entanglement allows qubits to be correlated with each other in ways that classical bits cannot be. This quantum correlation can enhance the performance of quantum algorithms for modeling more complex relationships and interactions. These quantum properties enable quantum computers to tackle problem-solving issues by handling them with enormous datasets and high-dimensional spaces more effectively than classical computers. This capability makes quantum computers and quantum algorithms ideal for applications in machine learning and optimization.

On the other hand, machine learning algorithms create models that learn and identify patterns from data in diverse domains, such as scientific research, technical applications, and industrial settings [4–6]. In the healthcare domain, machine learning algorithms have been widely employed for classifying biological data, identifying novel biomarkers, and exploring new disease subtypes.

Despite the significant success of traditional machine learning models, these approaches face challenges in handling large datasets and complex models with extensive computational operations. Dealing with large datasets can be time-consuming and expensive for them, limiting their scalability. Also, they may struggle with certain types of data with complex relationships and patterns, restricting their effectiveness. Quantum machine learning offers a promising solution by leveraging the unique properties of quantum systems to improve computational efficiency, handle complex data, and enhance scalability [7].

QML presents a significant potential to effectively influence biological and medical data analysis with the aim of minimizing the occurrence of medical mistakes [8]. Hence, utilizing the power of QML algorithms can enhance the analysis of medical data to improve the detection of early-stage diseases. In turn, it leads to enhancement of patient management, cost reduction, and ultimately, better treatment outcomes [9].

In an investigation focused on utilizing QML for classifying a particular type of heart disease, a comparison between quantum classifiers and classical algorithms was conducted using a cardiovascular dataset. In the study, it was claimed that the accuracy of the QML algorithms demonstrated an overall improvement [10]. One popular algorithm in QML for classification tasks is the Quantum Support Vector Machine (QSVM) which is considered a quantum counterpart of the classical Support Vector Machine (SVM) algorithm. QSVM due to its quantum computing capability has the potential to influence the classification metrics to provide better identification and diagnosis

In QSVM, quantum kernels are used for computational operations that are difficult to perform in related classical methods [11, 12]. The utilization of quantum kernels allows QSVM to explore the power of quantum computing in machine learning, particularly concerning kernel functions [13]. In addition, a kernel-based support vector machine was implemented to achieve higher accuracy compared to the classical SVM [14].

On real-world biological systems with their particular complexity and high dimensionality, a support vector machine with quantum kernel functions was employed to investigate the effect of feature engineering on classification accuracy [15]. For implementing quantum algorithms, it is essential to take into account the current quantum computational limitations. It means that the full potential of quantum methods is not universally accessible, and also the achieved results may not be generalizable across all systems. Hence, the predicted outcomes have only been observed in certain cases for demonstrating superior performance of quantum models compared to classical methods. For instance, in an investigation, the evaluation of various quantum machine learning techniques on five distinct datasets revealed that, on average, the QSVM algorithm consistently outperformed the classical counterparts in terms of accuracy by a few percentage points [16].

While, in another experiment on the Diabetes disease dataset, a comparative analysis was conducted to assess the performance of different QML models compared to a classical algorithm. The classical method demonstrated a slight superiority over the quantum algorithms with respect to overall accuracy [17].

In this research, we implemented the quantum support vector machine algorithm on biological high-dimensional real-world datasets to classify various cancer types and identify novel biomarkers. In comparison between the obtained results from quantum machine learning algorithms and the classical ones, the overall classification analysis showed that for noticeable number of datasets, the quantum model achieved superior results. Considering the limitation of quantum computers for the qubit numbers and the high dimensionality of gene expression data as features, we executed our QSVM algorithm on a classical quantum simulator.

2 Methods

In this section, we present our employed quantum machine learning method for classifying ten cancers and identifying novel biomarkers. We begin by providing an overview of the fundamental concepts of quantum computing. Next, we introduce quantum machine learning, which includes the use of quantum kernels. For comparing the results of our QSVM algorithm, we applied some other classical machine learning models including SVM (kernel='linear'), Random Forest (n_estimators: 100, min_samples_split: 2, bootstrap: True), Extreme Gradient Boosting (n_estimators: 100, max_depth: 6, learning_rate: 0.3, gamma: 0), Decision Tree (min_samples_split: 2, min_samples_leaf: 1, criterion: 'gini'), and Logistic Regression (C: 1.0, penalty: 'l2', solver: 'lbfgs', max_iter: 100, fit_intercept: True). Finally, we present and discuss the outcomes.

2.1 Quantum Computing

In classical computing, information is stored in bits as either 0 or 1. However, in quantum computing, the basic unit of information is stored as qubits, which are quantum bits. Mathematically, a qubit is represented as a vector in a two-dimensional complex Hilbert space, with the computational basis $|0\rangle$ and $|1\rangle$ [2]. Unlike classical bits, a qubit can exist in one of three states: $|0\rangle$, $|1\rangle$, or a superposition, which is a linear combination of the basis states, denoted as $c_0|0\rangle + c_1|1\rangle$. Quantum entanglement, a significant phenomenon in quantum mechanics, arises from the concept of superposition. It refers to the correlation between two or more quantum states, where each state can influence the others, even when physically separated by a great distance. Measurement of one entangled particle causes the collapse of its superposition state, instantaneously affecting the others. These quantum characteristics imply that entangled qubit states can store more information than individual ones. This property provides quantum computing with the potential for exponential speedup in solving traditionally difficult problems.

2.2 Quantum machine learning

Quantum machine learning combines quantum algorithms with classical machine learning methods to enhance computational complexity. QML algorithms consist of quantum classification, quantum clustering, quantum regression, quantum optimization, and several other models. These methods have developed with the aim of utilizing quantum computing principles to potentially outperform their classical counterparts in terms of speed and efficiency. However, these algorithms are still in the early stages of development and require further research and experimentation. Particularly, quantum classification algorithms seek to classify data points into different classes or categories. One of the key quantum classification algorithms is the QSVM, which uses quantum computing principles to find the optimal hyperplane that separates different classes in the feature space [18]. QSVM has the potential to outperform classical SVMs in certain cases by leveraging quantum parallelism and interference.

Various strategies have been developed to create QML algorithms, which serve as quantum counterparts to classical machine learning techniques [19]. Quantum-inspired machine learning techniques leverage quantum computing methods to improve classical machine learning algorithms [20]. Quantum-enhanced machine learning algorithms involve executing classical data on quantum computers [21]. The potential of quantum computers lies in their ability to utilize qubits, which provide an exponential computational space compared to classical computers operating in the Boolean space. Hybrid classical-quantum machine learning algorithms merge classical and quantum methods to enhance model performance [22]. In this research, we employ a quantum machine learning algorithm based on the support vector machine method to classify high-dimensional cancer data and identify biomarkers. Our approach falls into the category of hybrid classical-quantum machine learning algorithms, where we simulate the quantum algorithm on a classical computer.

2.3 Quantum support vector machine

Quantum support vector machine algorithm utilizes the benefits of quantum computing to improve the performance of classical SVM algorithms, which is a popular supervised machine learning algorithm commonly used for classification problems [23]. It operates by separating data points into the distinct classes through the identification of optimal

hyperplanes. The optimal hyperplane is determined by maximizing the margin between different classes, effectively ensuring the greatest distance from the nearest data points of each class. The SVM algorithms use kernels to classify the data that is not linearly separable. Kernel functions map data points into a higher-dimensional space for easier solving non-linearly separable problems using feature maps [24]. Depending on the data characteristics, some common kernel functions that are used in SVM include linear, polynomial, and radial basis function kernels. The introduction of quantum models into classical kernel methods gives rise to quantum kernels, which form the foundation of QSVM. In QML, quantum kernels are applied to map classical data into quantum states within a Hilbert space through the implementation of quantum feature maps [25].

In the QML framework, classical data points x from the input dataset χ are encoded into quantum states $|\psi(x)\rangle$ within a higher-dimensional Hilbert space ($\mathcal{H}$) during the quantum feature mapping process [26, 27]. Mathematically, this can be represented as:

$$\psi : \chi \rightarrow \mathcal{H}; x \in \chi, |\psi(x)\rangle \in \mathcal{H} \quad (1)$$

where ψ denotes a quantum feature encoding that employs a mathematical procedure for mapping data.

The quantum kernel counterparts of classical kernels are constructed based on the kernel function to build a quantum classifier. The kernel function κ calculates the inner product between two quantum states that represents two mapped data points x, x' in combination with a quantum feature map. It is defined as:

$$\kappa(x, x') = \langle \psi(x) | \psi(x') \rangle_{\mathcal{H}}; x, x' \in \chi \quad (2)$$

Here, $\langle \cdot | \cdot \rangle$ denotes the dot product [28]. By applying quantum kernels, the transformation of classical input data into quantum states can be represented through an encoding circuit [12]:

$$|\psi(x)\rangle = \mathcal{U}_{\psi}(x) |0^{\otimes n}\rangle; \mathcal{U}_{\psi}(x) = \prod_d \mathcal{U}_{\psi}(x) H^{\otimes n} \quad (3)$$

In this equation, $\mathcal{U}_{\psi}(x)$ indicates a quantum feature encoding circuit consisting of H (Hadamard gates) and $\mathcal{U}_{\psi}(x)$ (unitary operations containing Pauli gates). The variable n denotes the number of qubits used for encoding, while d represents the depth of the circuit for creating the quantum feature map [12]. The initial state for n qubits is $|0^{\otimes n}\rangle = |0 \dots 0\rangle$. The quantum computing process involves modifying qubits using quantum gates, which integrate traditional logical operators and gates. Quantum gates, such as Hadamard gates and controlled Pauli gates, serve as the fundamental components of quantum circuits. Hadamard gates facilitate the transition of qubits into a superposition state, while controlled Pauli gates enable rotations of qubits around the x , y , and z axes, thereby facilitating quantum entanglement among qubits [16]. To estimate an observable value within a quantum circuit, multiple measurements are performed, and the resultant output is utilized as a prediction [29].

2.4 Our approach

In our research, we utilized the QML scikit-qulacs package [30] to classify different types of cancers and identify biomarkers. This quantum machine learning algorithm is based on Qulacs [31], one of the fastest quantum circuit simulators. Qulacs is designed to assist researchers in quantum computing operations and has demonstrated significant speed-up in exploring and performing algorithms with high accuracy. The scikit-qulacs package introduces a quantum machine learning algorithm that utilizes the quantum circuit simulator Qulacs. In this QML algorithm, quantum circuits are created using the "Learning Circuit" function from the "skqulacs.circuit" sub-package in the "skqulacs" package [32]. A quantum circuit, represented as an array of quantum gates, is constructed using the "ParametricQuantumCircuit" class. The "create_yzcx_ansatz" function from the "skqulacs.circuit.pre_defined" sub-package is used to generate quantum gates. The resulting circuit can be visualized using the "circuit_drawer" function from the "qulacsvis" package [33]. The classification process involves quantum support vector classification, which is executed using the "QSVM" function from the "skqulacs.qsvm" package [34]. We implemented the "skqulacs.qsvm" package in our data to take advantage of its capabilities in performing classification.

Furthermore, we employed the Fisher score algorithm for feature selection. This algorithm is a dimensional reduction and feature-ranking approach commonly used for selecting genes from expression profile data. The procedure for feature selection is as follows: Given $X \in \mathbb{R}_{m \times n}$, a matrix of gene expression data (where m and n represent the number

of genes and samples, respectively), and $NG = (U, C, D, \delta)$, a neighborhood decision system for gene expression data, the Fisher score is calculated as:

$$f(Z) = \frac{\text{tr}(A_b)}{\text{tr}(A_w)} \quad (4)$$

Here, $\text{tr}()$ denotes the trace of a matrix, A_w represents the scatter matrix within the same class, and A_b represents the scatter matrix between cancer samples and their paired normal samples. An exploratory strategy is commonly employed to compute the score for each gene using similar criteria. The Fisher score for the k -th gene is then obtained as:

$$f(i) = \frac{\sum_{i=1}^C n_i (\mu_i^k - \mu^k)^2}{\sum_{i=1}^C n_i (\sigma_i^k)^2} \quad (5)$$

Here, n_i refers to the sample number of the i -th class, μ_i^k and σ_i^k represent the mean and standard deviation of the samples from the i -th category with the k -th gene, respectively, and μ^k is the mean magnitude of the k -th gene samples [35, 36]. We wrote and executed the code in a Python 3.11.3 environment.

3 Data collection

We downloaded GDC TCGA RNA-seq (HTSeq) counts of 10 cancers including Bladder Cancer (BLCA), Colon Adenocarcinoma (COAD), Head and Neck Squamous Cell Carcinoma (HNSC), Kidney Renal Clear Cell Carcinoma (KIRC), Liver hepatocellular carcinoma (LIHC), Lung Adenocarcinoma (LUAD), Prostate Adenocarcinoma (PRAD), Rectal Adenocarcinoma (READ), Stomach Adenocarcinoma (STAD), and Thyroid Cancer (THCA) from UCSC Xena database (<https://xenabrowser.net/>). A total of 4785 tumor and healthy samples have been employed. The details of the samples are described in Table 1.

The Ensemble IDs were mapped to official gene symbols. The Ensemble IDs without gene names were excluded. The database contains logarithm-transformed data. For this reason, before starting the gene filtering, we converted them back to integer raw counts.

3.1 Gene filtering and determining differentially expressed genes

In the data preprocessing stage, we applied several methods for gene filtering and determining differentially expressed genes (DEGs). First, we removed all rows with a zero-sum. Next, we discarded genes with variances less than the 1st quartile. Subsequently, we performed TMM normalization and $\log_2$ transformation on the data. To identify DEGs, we employed the edgeR and limma voom methods using the edgeR (version 3.36.0) and limma (3.50.0) packages in the R programming environment (version 4.3.1), respectively. The criteria we considered were a Benjamini-Hochberg adjusted p -value < 0.05 [37] and $|\log_2 FC| > 3$. Finally, we identified the common DEGs identified by both methods for further analysis. We excluded long non-coding RNAs from the analysis.

Table 1 Details of the cancer datasets

Cancer	# Tumor samples	# Paired normal samples	# Train samples	# Train samples after SMOTE	# Test samples
BLCA	411	19	301	578	129
COAD	468	41	356	662	153
HNSC	500	44	380	696	164
KIRC	534	72	424	746	182
LIHC	371	50	294	518	127
LUAD	524	59	408	736	175
PRAD	498	52	385	706	165
READ	166	10	123	230	53
STAD	375	32	284	522	123
THCA	501	58	391	714	168

3.2 Preparing data for classification

To prepare the data for classification, we randomly partitioned the dataset into train and test sets using Python 3.11.3 in a 70/30 ratio. To address the imbalance between normal and cancer classes in the training dataset, we employed Synthetic Minority Over-Sampling Technique (SMOTE). SMOTE generates synthetic data points in the feature space by utilizing the K-Nearest Neighbor (KNN) algorithm [38]. The number of train and test samples have mentioned in Table 1.

4 Results and discussion

In our study, we performed some classical models including classical SVM (CSVM), RF, DT, XGBoost, and LR, as well as QSVM algorithm to classify ten types of cancer and identify biomarkers. The analysis was simulated on a classical computer. The following sections provide a detailed explanation of the analysis steps and present the obtained results.

4.1 Determination of DEGs

Table 2 presents the number of genes after undergoing various filtering steps (as described in the Data Collection section), as well as the number of DEGs identified by edgeR, limma, and the common DEGs identified by both methods (Supplementary data file 1). For further analysis, we focused on the shared DEGs as a strict criterion for selecting genes to be included in the machine learning analysis step.

Table 2 Number of genes after filtering and number of DEGs

Cancer	# Genes	# DEGs_edgeR	# DEGs_limma	# DEGs_ common
BLCA	6550	337	249	146
COAD	5575	314	305	180
HNSC	5792	325	224	154
KIRC	5761	370	392	239
LIHC	4842	233	217	64
LUAD	6675	478	252	151
PRAD	4876	88	23	12
READ	4344	229	231	154
STAD	5858	303	162	86
THCA	4430	177	150	92

Table 3 The top 5 ranked genes based on their Fisher scores

Cancer type	BLCA	COAD	HNSC	KIRC	LIHC	LUAD	PRAD	READ	STAD	THCA
Features	CMTM3	CDH3	MMP11	AQP2	MARCO	FHL1	CLCA2	CDH3	MYOC	LRP4
	PRKCE	CEMIP	CA9	UMOD	OIT3	CAV1	DLX1	GRIN2D	COL10A1	GABRB2
	RRM2	KRT80	ADIPOQ	MUC15	FCN3	FABP4	SLC45A2	FOXQ1	ACAN	KLHDC8A
	EEF1A1P5	ETV4	ADH1B	HS6ST2	CRHBP	PYCR1	PCA3	KRT80	CLEC3B	METTL7B
	RPLP06	OTOP2	SH3BGRL2	SLC9A4	CLEC4G	AGER	AMACR	ETV4	CST1	PRR15

4.2 Feature selection and performance analysis of CSVM and QSVM

The Fisher score, a supervised feature selection approach, was used to rank the genes based on their importance. We selected the top 5 ranked genes based on their Fisher scores as indicated in Table 3. The expression amount of each sample served as input data for both classical and quantum machine learning methods. The bold ones are new biomarkers for cancer diagnosis.

For QSVM, we also applied five qubits according to the top features ranked by the Fisher score. We also applied different depth layers (1-20), and selected the ones with the higher obtained accuracy. Figure 1 demonstrates the constructed circuits for 1, 3, 4 layers as examples.

We used parameterized quantum circuit (PQC) structure to demonstrate the quantum circuit for each cancer data. In PQC, we used YZ-CX circuit constructions consisting of single-qubit y and z rotations and two-qubit entangling gates. Each quantum circuit is defined with N qubits that are used for implementing quantum machine learning algorithms. Depending on the cost of applying each element, we defined every circuit with *l* layers that show the repetition of the circuit's basic block. Table 4 and Fig. 2 present the test accuracy, precision, recall, F1-score, and AUC, while Figure S1 illustrates the receiver operating characteristic (ROC) curves generated by classical machine learning algorithms and QSVM.

The results show that LR and RF leads to the better classification performances among classical machine learning algorithms. Logistic regression, while is a simpler linear model, has been widely utilized in cancer research due to its interpretability and ease of implementation [39]. Studies have shown that it can achieve comparable results to more complex models in specific scenarios, particularly in low-dimensional datasets [40]. However, its limitations become evident in high-dimensional settings, where it may struggle to capture the intricate relationships between features effectively [41]. As we firstly employed the feature selection Fisher approach, it could effectively classify the cancer datasets. On the other hand, the ensemble nature of RF allows it to manage high feature-to-sample ratios effectively, making it particularly suitable for cancer classification tasks where the data can be complex and noisy [41].

The classification reports also demonstrate the good classification performance of the QSVM-Kernel method. The results also illustrate that our quantum-integrated workflow provides a clear advantage over classical algorithms in terms of accuracy when applied to identical datasets, particularly in cases including HNSC, KIRC, LIHC, LUAD, READ, and STAD. The larger AUC indicates that the quantum support vector machine (QSVM) demonstrates superior performance for COAD, HNSC, LIHC, PRAD, READ, THCA, and STAD. While previous studies have provided valuable insights into the application of quantum models for cancer classification, there are still some limitations such as the number of qubits that need to be addressed. In one study [42], a hybrid quantum-mechanical system with two qubits was employed to encode, process, and classify images of cancerous and non-cancerous pigmented skin lesions. This approach demonstrated the

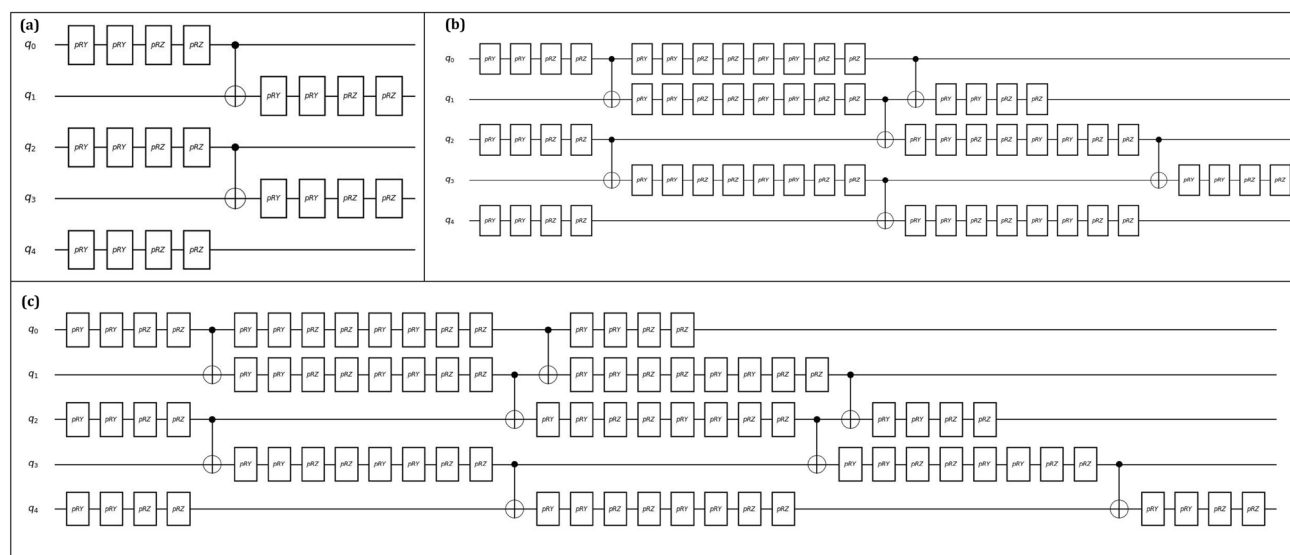


Fig. 1 The circuit constructed by five qubits and **a** 1 and **b** 3, and **c** 4 layers

Table 4 The test accuracy, precision, recall, F1-score, and ACU obtained by classic ML algorithms and QSVM The terms “Ln” denote the applied depth layer, in which “n” represent the number of layer

Classification method	Accuracy	Precision	Recall	F1-score	AUC
SVM	BLCA: 0.97	BLCA: 0.78	BLCA: 0.89	BLCA: 0.83	BLCA: 1.00
	COAD: 0.99	COAD: 0.93	COAD: 0.99	COAD: 0.96	COAD: 0.90
	HNSC: 0.99	HNSC: 0.93	HNSC: 0.99	HNSC: 0.96	HNSC: 0.99
	KIRC: 0.96	KIRC: 0.90	KIRC: 0.98	KIRC: 0.94	KIRC: 0.98
	LIHC: 0.96	LIHC: 0.90	LIHC: 0.92	LIHC: 0.91	LIHC: 0.92
	LUAD: 0.98	LUAD: 0.91	LUAD: 0.99	LUAD: 0.95	LUAD: 0.99
	PRAD: 0.94	PRAD: 0.83	PRAD: 0.97	PRAD: 0.88	PRAD: 0.97
	READ: 1.00	READ: 1.00	READ: 1.00	READ: 1.00	READ: 1.00
	STAD: 1.00	STAD: 1.00	STAD: 1.00	STAD: 1.00	STAD: 1.00
	THCA: 0.95	THCA: 0.89	THCA: 0.94	THCA: 0.91	THCA: 0.94
RF	BLCA: 0.99	BLCA: 1.00	BLCA: 0.90	BLCA: 0.94	BLCA: 0.90
	COAD: 1.00	COAD: 1.00	COAD: 1.00	COAD: 1.00	COAD: 1.00
	HNSC: 0.99	HNSC: 0.93	HNSC: 0.99	HNSC: 0.96	HNSC: 0.99
	KIRC: 0.99	KIRC: 0.98	KIRC: 1.00	KIRC: 0.99	KIRC: 1.00
	LIHC: 0.97	LIHC: 0.92	LIHC: 0.92	LIHC: 0.92	LIHC: 0.92
	LUAD: 0.98	LUAD: 0.97	LUAD: 0.94	LUAD: 0.95	LUAD: 0.94
	PRAD: 0.96	PRAD: 0.90	PRAD: 0.94	PRAD: 0.92	PRAD: 0.94
	READ: 1.00	READ: 1.00	READ: 1.00	READ: 1.00	READ: 1.00
	STAD: 0.98	STAD: 0.99	STAD: 0.89	STAD: 0.93	STAD: 0.89
	THCA: 0.98	THCA: 0.97	THCA: 0.93	THCA: 0.95	THCA: 0.93
XGBoost	BLCA: 0.98	BLCA: 0.99	BLCA: 0.80	BLCA: 0.87	BLCA: 0.80
	COAD: 1.00	COAD: 1.00	COAD: 1.00	COAD: 1.00	COAD: 1.00
	HNSC: 0.98	HNSC: 0.92	HNSC: 0.95	HNSC: 0.94	HNSC: 0.95
	KIRC: 0.99	KIRC: 0.98	KIRC: 1.00	KIRC: 0.99	KIRC: 1.00
	LIHC: 0.94	LIHC: 0.91	LIHC: 0.76	LIHC: 0.82	LIHC: 0.76
	LUAD: 0.98	LUAD: 0.99	LUAD: 0.92	LUAD: 0.95	LUAD: 0.92
	PRAD: 0.95	PRAD: 0.90	PRAD: 0.86	PRAD: 0.88	PRAD: 0.86
	READ: 0.98	READ: 0.83	READ: 0.99	READ: 0.90	READ: 0.99
	STAD: 0.98	STAD: 0.99	STAD: 0.89	STAD: 0.93	STAD: 0.89
	THCA: 0.97	THCA: 0.95	THCA: 0.93	THCA: 0.94	THCA: 0.93
DT	BLCA: 0.97	BLCA: 0.79	BLCA: 0.79	BLCA: 0.79	BLCA: 0.79
	COAD: 1.00	COAD: 1.00	COAD: 1.00	COAD: 1.00	COAD: 1.00
	HNSC: 0.98	HNSC: 0.92	HNSC: 0.95	HNSC: 0.94	HNSC: 0.95
	KIRC: 0.97	KIRC: 0.90	KIRC: 0.98	KIRC: 0.94	KIRC: 0.98
	LIHC: 0.95	LIHC: 0.91	LIHC: 0.86	LIHC: 0.88	LIHC: 0.86
	LUAD: 0.98	LUAD: 0.96	LUAD: 0.92	LUAD: 0.94	LUAD: 0.92
	PRAD: 0.89	PRAD: 0.75	PRAD: 0.83	PRAD: 0.78	PRAD: 0.83
	READ: 1.00	READ: 1.00	READ: 1.00	READ: 1.00	READ: 1.00
	STAD: 0.98	STAD: 0.93	STAD: 0.88	STAD: 0.91	STAD: 0.88
	THCA: 0.96	THCA: 0.93	THCA: 0.93	THCA: 0.93	THCA: 0.93
LR	BLCA: 0.99	BLCA: 1.00	BLCA: 0.90	BLCA: 0.94	BLCA: 0.90
	COAD: 1.00	COAD: 1.00	COAD: 1.00	COAD: 1.00	COAD: 1.00
	HNSC: 0.99	HNSC: 0.93	HNSC: 0.99	HNSC: 0.96	HNSC: 0.99
	KIRC: 0.99	KIRC: 0.98	KIRC: 1.00	KIRC: 0.99	KIRC: 1.00
	LIHC: 0.97	LIHC: 0.92	LIHC: 0.92	LIHC: 0.92	LIHC: 0.92
	LUAD: 0.99	LUAD: 0.97	LUAD: 1.00	LUAD: 0.99	LUAD: 1.00
	PRAD: 0.97	PRAD: 0.94	PRAD: 0.92	PRAD: 0.93	PRAD: 0.92
	READ: 1.00	READ: 1.00	READ: 1.00	READ: 1.00	READ: 1.00
	STAD: 0.98	STAD: 0.99	STAD: 0.89	STAD: 0.93	STAD: 0.89
	THCA: 0.96	THCA: 0.94	THCA: 0.91	THCA: 0.92	THCA: 0.91
QSVM	BLCA: L13: 0.98	BLCA: 0.83	BLCA: 0.89	BLCA: 0.86	BLCA: 0.89
	COAD: L1: 0.99	COAD: 0.93	COAD: 0.99	COAD: 0.96	COAD: 1.00
	HNSC: L3: 0.99	HNSC: 0.93	HNSC: 0.99	HNSC: 0.96	HNSC: 0.99
	KIRC: L17: 0.99	KIRC: 0.96	KIRC: 0.99	KIRC: 0.97	KIRC: 0.99
	LIHC: L19: 0.97	LIHC: 0.91	LIHC: 0.95	LIHC: 0.93	LIHC: 0.95
	LUAD: L8: 0.99	LUAD: 0.95	LUAD: 0.99	LUAD: 0.97	LUAD: 0.99
	PRAD: L1: 0.95	PRAD: 0.84	PRAD: 0.97	PRAD: 0.89	PRAD: 0.97
	READ: L1: 1.00	READ: 1.00	READ: 1.00	READ: 1.00	READ: 1.00
	STAD: L1: 1.00	STAD: 1.00	STAD: 1.00	STAD: 1.00	STAD: 1.00
	THCA: L4: 0.97	THCA: 0.93	THCA: 0.95	THCA: 0.94	THCA: 0.95

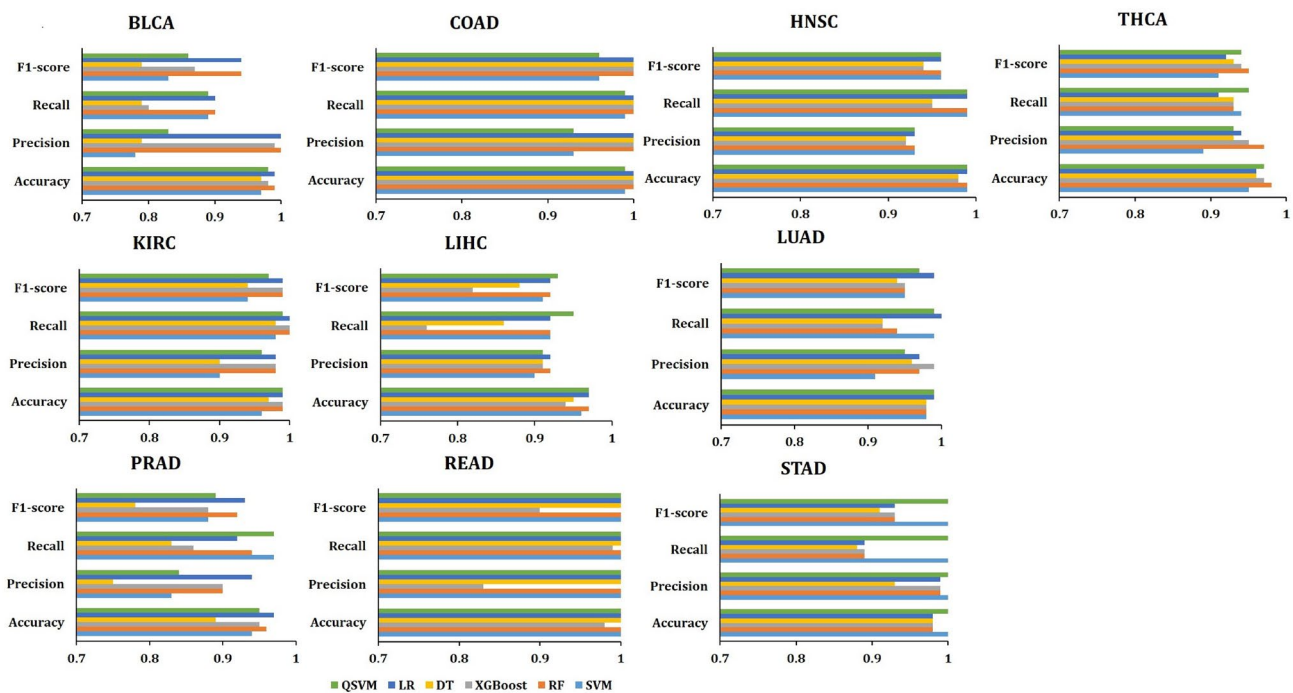


Fig. 2 The test accuracy, precision, recall, and F1-score obtained by classic ML algorithms and QSVM

potential of quantum machine learning in image-based cancer classification. Furthermore, researchers used the QSVM model to distinguish between malignant breast cancer tumors and non-cancerous benign tumors [43]. They applied principal component analysis for feature reduction, taking into account the limited number of available qubits. However, we provided a stepwise approach containing a feature selection method and then applied QSVM with a negotiable number of qubits. It led to the finding of novel biomarkers along with confirmation of the previously identified biomarkers.

Essentially, different types of cancer can be considered real-world systems whose structures consist of a large number of variety parameters. This means that classifying these high-dimensional systems is complicated and time-consuming on classical computers [44–46]. While conventional computers can handle cancer classification tasks, quantum computers have the potential to solve complex high-dimensional problems more efficiently. Quantum computers, due to their unique capabilities such as superposition and entanglement, offer advantages over traditional computers in certain cases, particularly for cancer type classification. One key advantage of quantum computers for cancer type classification is their ability to handle high-dimensional data more effectively than classical computers. Quantum SVMs can represent data in high-dimensional spaces efficiently, which is crucial for analyzing complex cancer genomics datasets [47]. Furthermore, quantum SVMs have shown potential for achieving higher classification accuracy in shorter training times compared to classical SVMs, especially for non-linearly separable datasets. The quantum kernel methods used in QSVM can lead to better performance than classical kernels in certain tasks. As quantum hardware continues to improve, there is increasing potential for quantum SVMs to outperform classical SVMs for specific cancer classification tasks. Quantum advantage has already been demonstrated in simulations and on real quantum devices using error mitigation techniques.

Additionally, quantum computers can handle larger datasets more effectively than classical computers due to their parallelization capabilities. This is particularly relevant for cancer classification, where large genomic and imaging datasets are becoming increasingly available. QML models are designed to classify different cancer types and identify biomarkers applicable in various real-life situations [44, 46]. QML techniques can enhance personalized treatment plans by analyzing individual patient data to predict their cancer type and how they might respond to specific treatments. By identifying unique biomarkers, healthcare providers can tailor therapies that improve treatment efficacy and minimize side effects [48]. Moreover, effective classification of cancer types aids in early detection. QML models can analyze imaging data or genetic profiles to identify anomalies indicating cancer [42, 46], facilitating timely interventions that are crucial for successful treatment. Throughout the treatment process, professionals can utilize the analysis

of patient data over time to assess how well a cancer is responding to treatment. Furthermore, healthcare institutions can leverage QML models to analyze large datasets from populations to identify trends in cancer types and treatment responses. In drug development, QML can identify biomarkers that predict how different cancers respond to new drugs. This can streamline the drug discovery process [49], allowing researchers to focus on compounds that are more likely to be effective for specific cancer types. Additionally, QML models can be integrated with other technologies, such as electronic health records [10] to provide a holistic view of patient health, resulting in better management of disease care.

4.3 Promoter DNA methylation status and feature's expression

Promoter DNA methylation is an epigenetic modification that regulates gene expression and often occurs early in tumorigenesis [50]. In our study, we examined the promoter DNA methylation status of the classifier genes and visualized the data using box-whisker plots. To assess the significance of expression level differences between normal and tumor groups, we employed Welch's T-test.

Figures S2–S11 depict the methylation levels of features that showed the relationship with the expression values. The observed changes in promoter DNA methylation may provide potential mechanisms underlying the downregulation or upregulation of classifier gene expression as determined by QSVM.

The significant declined promotor DNA methylation levels of PRM2 and PRKCE in BLCA, KRT80 in COAD, ADH1B, ADIPOQ, and CA9 in HNSC, HS6ST2 in KIRC, CLEC4G, CRHBP, FCN3, MARCO in LIHC, CLCA2, AMACR, and SLC45A2 in PRAD, FOXQ1, ETV4, and KRT80 in READ, COL10A1 in STAD, LRP4, KLHDC8A, METTL7B, and PRR15 in THCA accompany with increased gene expression as well as significantly increased promotor DNA methylation level of CMTM3 in BLCA, CDH3, ETV4, and OTOP2 in COAD, AQP2, MUC15, and SLC9A4 in KIRC, CAV1 and FHL1 in LUAD, GRIN2D and CDH3 in READ, and DLX1 in PRAD were observed.

5 Conclusion

In the present study, we utilized a quantum machine learning algorithm for the classification of high-dimensional RNA-seq cancer data. Our findings demonstrate that the results obtained from the quantum machine learning approach are consistent with those obtained from classical machine learning algorithms, indicating the applicability of quantum methods for analyzing high-throughput biological data in certain cases. Additionally, our analysis successfully identified novel biomarkers. These findings highlight the potential of quantum machine learning in medical research and the discovery of valuable biomarkers.

Author contributions M-ZG performed the bioinformatics analysis. M-ZG and EA performed the QML analysis, interpreters the results, and wrote the manuscript.

Funding No funding.

Data availability The RNA-seq data expression were downloaded from UCSC Xena database (<https://xenabrowser.net/>). The codes are available in https://github.com/Mohadesehzareei/skulacs_QSVM.

Declarations

Conflict of interest The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Preskill J. Quantum computing in the NISQ era and beyond. *Quantum*. 2018;2:79.
2. Preskill J. Lecture notes for physics 229: quantum information and computation. California Inst Technol. 1998;16(1):1–8.
3. Gill SS, Kumar A, Singh H, Singh M, Kaur K, Usman M, Buyya R. Quantum computing: a taxonomy, systematic review and future directions. *Softw Pract Exp*. 2022;52(1):66–114.
4. Jordan MI, Mitchell TM. Machine learning: trends, perspectives, and prospects. *Science*. 2015;349(6245):255–60.
5. Sarker I. Machine learning: algorithms, real-world applications and research directions. *SN Comput Sci*. 2021;2(3):160.
6. Wuest T, Weimer D, Irgens C, Thoben K-D. Machine learning in manufacturing: advantages, challenges, and applications. *Prod Manuf Res*. 2016;4(1):23–45.
7. Mironowicz P, Mandarino A, Yilmaz A, Ankenbrand T. Applications of quantum machine learning for quantitative finance. *arXiv preprint arXiv:240510119* 2024.
8. Maheshwari D, Garcia-Zapirain B, Sierra-Sosa D. Quantum machine learning applications in the biomedical domain: a systematic review. *IEEE Access*. 2022;29(10):80463–84.
9. Ullah U, Garcia-Zapirain B. Quantum machine learning revolution in healthcare: a systematic review of emerging perspectives and applications. *IEEE Access*. 2024;12:11423–50.
10. Maheshwari D, Ullah U, Marulanda PAO, Jurado AG-O, Gonzalez ID, Merodio JMO, Garcia-Zapirain B. Quantum machine learning applied to electronic healthcare records for ischemic heart disease classification. *Hum-Cent Comput Inf Sci*. 2023;13(06).
11. Rebentrost P, Mohseni M, Lloyd S. Quantum support vector machine for big data classification. *Phys Rev Lett*. 2014;113(13): 130503.
12. Havlíček V, Córcoles AD, Temme K, Harrow AW, Kandala A, Chow JM, Gambetta JM. Supervised learning with quantum-enhanced feature spaces. *Nature*. 2019;567(7747):209–12.
13. Park S, Park DK, Rhee JK. Variational quantum approximate support vector machine with inference transfer. *Sci Rep*. 2023;13(1):3288.
14. Willsch D, Willsch M, De Raedt H, Michielsen K. Support vector machines on the D-Wave quantum annealer. *Comput Phys Commun*. 2020;248: 107006.
15. Vasques X, Paik H, Cif L. Application of quantum machine learning using quantum kernel algorithms on multiclass neuron M-type classification. *Sci Rep*. 2023;13(1):11541.
16. Simões RDM, Huber P, Meier N, Smailov N, Fuchsli RM, Stockinger K. Experimental evaluation of quantum machine learning algorithms. *IEEE Access*. 2023;11:6197–208.
17. Maheshwari D, Sierra-Sosa D, Garcia-Zapirain B. Variational quantum classifier for binary classification: real vs synthetic dataset. *IEEE Access*. 2021;10:3705–15.
18. Tychola KA, Kalampokas T, Papakostas GA. Quantum machine learning—an overview. *Electronics*. 2023;12(11):2379.
19. Zeguendry A, Jarir Z, Quafafou M. Quantum machine learning: a review and case studies. *Entropy*. 2023;25(2):287.
20. Felser T, Trenti M, Sestini L, Gianella A, Zuliani D, Lucchesi D, Montangero S. Quantum-inspired machine learning on high-energy physics data. *npj Quantum Inform*. 2021;7(1):111.
21. Dunjko V, Taylor JM, Briegel HJ. Quantum-enhanced machine learning. *Phys Rev Lett*. 2016;117(13): 130501.
22. Adhikary S, Dangwal S, Bhowmik D. Supervised learning with a quantum classifier using multi-level systems. *Quantum Inform Process*. 2020;19:1–12.
23. Afsaneh E, Sharifdini A, Ghazzaghi H, Ghobadi MZ. Recent applications of machine learning and deep learning models in the prediction, diagnosis, and management of diabetes: a comprehensive review. *Diabetol Metabol Syndr*. 2022;14(1):1–39.
24. Huang H-Y, Broughton M, Mohseni M, Babbush R, Boixo S, Neven H, McClean JR. Power of data in quantum machine learning. *Nat Commun*. 2021;12(1):2631.
25. Hanco-Quispe JK, Borda-Colque JP, Torres-Cruz F. Quantum machine learning applied to the classification of diabetes. *arXiv preprint arXiv:00109* 2022.
26. Goto T, Tran QH, Nakajima K. Universal approximation property of quantum machine learning models in quantum-enhanced feature spaces. *Phys Rev Lett*. 2021;127(9): 090506.
27. Jerbi S, Fiderer LJ, Poulsen Nautrup H, Kübler JM, Briegel HJ, Dunjko V. Quantum machine learning beyond kernel methods. *Nat Commun*. 2023;14(1):517.
28. Schuld M, Killoran N. Quantum machine learning in feature Hilbert spaces. *Phys Rev Lett*. 2019;122(4): 040504.
29. Schuld M, Sweke R, Meyer JJ. Effect of data encoding on the expressive power of variational quantum-machine-learning models. *Phys Rev A*. 2021;103(3): 032430.
30. [<https://qulacs-osaka.github.io/scikit-qulacs/index.html>]
31. Suzuki Y, Kawase Y, Masumura Y, Hiraga Y, Nakadai M, Chen J, Nakanishi KM, Mitarai K, Imai R, Tamiya S. Qulacs: a fast and versatile quantum circuit simulator for research purpose. *Quantum*. 2021;5:559.
32. [<https://qulacs-osaka.github.io/scikit-qulacs/skqulacs.html>]
33. https://qulacs-osaka.github.io/scikit-qulacs/notebooks/circuit_visualize.html. In.
34. <https://qulacs-osaka.github.io/scikit-qulacs/skqulacs.qsvm.qsvc.html>. In.
35. Sun L, Zhang X-Y, Qian Y-H, Xu J-C, Zhang S-G, Tian Y. Joint neighborhood entropy-based gene selection method with fisher score for tumor classification. *Appl Intell*. 2019;49:1245–59.
36. Li C, Xu J. Feature selection with the Fisher score followed by the maximal clique centrality algorithm can accurately identify the hub genes of hepatocellular carcinoma. *Sci Rep*. 2019;9(1):17283.
37. Ghobadi MZ, Afsaneh E, Emamzadeh R, Soroush M. Potential miRNA-gene interactions determining progression of various ATLL cancer subtypes after infection by HTLV-1 oncovirus. *BMC Med Genom*. 2023;16(1):1–9.
38. Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP. SMOTE: synthetic minority over-sampling technique. *J Artif Intell Res*. 2002;16:321–57.
39. Teixeira M, Silva F, Ferreira RM, Pereira T, Figueiredo C, Oliveira HP. A review of machine learning methods for cancer characterization from microbiome data. *NPJ Precis Oncol*. 2024;8(1):123.

40. Houfani D, Slatnia S, Kazar O, Remadna I, Saouli H, Ortiz G, Merizig A. An improved model for breast cancer diagnosis by combining PCA and logistic regression techniques. *Int J Comput Digit Syst.* 2023;13(1):701–16.
41. Couronné R, Probst P, Boulesteix A-L. Random forest versus logistic regression: a large-scale benchmark experiment. *BMC Bioinformatics.* 2018;19(1):270.
42. Iyer V, Ganti B, Hima Vyshnavi AM, Krishnan Namboori PK, Iyer S. Hybrid quantum computing based early detection of skin cancer. *J Interdiscip Math.* 2020;23(2):347–55.
43. Vashisth S, Dhall I, Aggarwal G. Design and analysis of quantum powered support vector machines for malignant breast cancer diagnosis. *J Intell Syst.* 2021;30(1):998–1013.
44. Flöther FF. The state of quantum computing applications in health and medicine. *Res Direct Quantum Technol.* 2023;1: e10.
45. Maheshwari D, Garcia-Zapirain B, Sierra-Sosa D. Quantum machine learning applications in the biomedical domain: a systematic review. *IEEE Access.* 2022;10:80463–84.
46. Khan MA-Z, Innan N, Galib AAO, Bennai M: Brain tumor diagnosis using quantum convolutional neural networks. *arXiv preprint arXiv:240115804* 2024.
47. Park J-E, Quanz B, Wood S, Higgins H, Harishankar R: Practical application improvement to quantum SVM: theory to practice. *arXiv preprint arXiv:201207725* 2020.
48. Ghobadi MZ, Emamzadeh R, Teymoori-Rad M, Afsaneh E. Exploration of blood–derived coding and non-coding RNA diagnostic immunological panels for COVID-19 through a co-expressed-based machine learning procedure. *Front Immunol.* 2022;13:1001070.
49. Wong YK, Zhou Y, Liang YS, Qiu H, Wu YX, He B. The new answer to drug discovery: quantum machine learning in preclinical drug development. In *IEEE 4th International Conference on Pattern Recognition and Machine Learning (PRML)*. 2023;557–64.
50. Bouras E, Karakioulaki M, Bougioukas KI, Aivaliotis M, Tzimagiorgis G, Chourdakis M. Gene promoter methylation and cancer: an umbrella review. *Gene.* 2019;710:333–40.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.